Newton School of Technology

Data Visualization & Analytics — DVA Capstone 2

Final Project Report

Swiggy Restaurant Analysis

Sector: Food Tech / Quick Commerce

Team ID	G-4 | Section D
Project Title	Swiggy Restaurant Analysis
Sector	Food Tech / Quick Commerce
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Submission Date	April 27, 2026
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1. Executive Summary

Problem

India's food delivery market is intensely competitive, with Swiggy serving millions of orders across dozens of cities. Restaurant operators and platform strategists lack a consolidated view of what drives customer ratings, how pricing varies across geographies, and which food categories generate the most demand. This project addresses that gap by analysing 91,788 cleaned order records from January to August 2025 across 28 Indian cities.

Approach

A fully automated Python ETL pipeline (extraction → cleaning → EDA → statistical analysis → final load) was built and executed across five modular notebooks. The cleaned dataset was then loaded into Tableau to build an interactive two-view dashboard — an Executive Overview and a Segmentation view — enabling decision-makers to explore KPIs by city, price tier, and time period.

Key Insights

- Bengaluru dominates order volume (~10,300 orders), nearly doubling the next-highest cities, indicating a strong first-mover and density advantage for Swiggy in Karnataka.
- KFC and McDonald's lead all restaurants by order count (7,700+ and 7,100+ respectively), reflecting the outsized demand for QSR chains on food delivery platforms.
- Mid-range dishes (₹150–₹400) account for 54,246 orders — the largest price segment — confirming that Indian consumers favour value-for-money options over budget or premium tiers.
- Ratings are statistically significantly different across cities and categories (ANOVA $p < 0.05$), but price has near-zero correlation with rating (Spearman $r = 0.03$), disproving the assumption that costlier food is rated higher.

Key Recommendations

- Invest in demand replication from Bengaluru to Tier-2 cities by identifying the restaurant mix and category portfolio that drives engagement there.
- Prioritise mid-range category expansion and promotions since this is where consumer spending is most concentrated.
- Use city-level rating differences as a quality signal to identify underperforming markets and trigger targeted restaurant partner interventions.
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2. Sector & Business Context

India's online food delivery sector is one of the fastest-growing segments of the consumer internet economy. With platforms like Swiggy and Zomato processing tens of millions of orders monthly, food delivery has moved from an urban convenience to a mainstream consumption channel across Tier-1 and Tier-2 cities alike.

Key industry challenges include: intense pricing pressure from rival platforms and restaurant aggregators; the difficulty of maintaining consistent food quality and customer ratings at scale; geographic fragmentation in food preferences; and the growing complexity of managing thousands of restaurant partners with highly varied menus and price points.

The primary decision-maker this project serves is the Swiggy Category & Growth team — analysts and managers who determine which cities and restaurant categories to invest in, which promotional bundles to offer, and where to intervene on quality. Secondary users include restaurant partner success managers who need city-level benchmarks.

The problem was chosen because raw order and rating data from Swiggy encodes valuable signals — about city-level demand concentration, pricing strategy effectiveness, and category preferences — that are not visible without systematic analysis. The business value of solving it is twofold: better allocation of marketing spend across geographies, and stronger data-driven conversations with restaurant partners about pricing and quality.

3. Problem Statement & Objectives

Formal Problem Definition

Given Swiggy's order dataset spanning January–August 2025 across 28 Indian cities, analyse the distribution and drivers of restaurant ratings, pricing patterns, demand concentration, and temporal order trends to surface actionable insights for platform growth and restaurant partner strategy.

Project Scope

- In scope: Descriptive EDA, statistical hypothesis testing, KPI computation, and Tableau dashboard visualisation of the provided dataset.
- Out of scope: Real-time data ingestion, customer-level behaviour analysis, delivery logistics metrics, and predictive/ML modelling.

Objectives

- Identify top-performing cities, states, restaurants, and food categories by order volume.
- Analyse the distribution of pricing and ratings across the platform.
- Test whether ratings differ significantly across cities and categories using ANOVA.
- Measure the relationship between price and rating using Spearman correlation.
- Deliver an interactive Tableau dashboard supporting executive-level and operational decision-making.

Success Criteria

- A fully automated ETL pipeline with reproducible outputs committed to GitHub.
- Statistical tests with documented hypotheses, results, and interpretations.
- A publicly accessible Tableau dashboard with at least two views and interactive filters.
- A 10–15 page academic report mapping every insight to a business recommendation.

4. Data Description

Source: Kaggle — Swiggy Food Orders Dataset. The raw file swiggy_data_raw.csv contains 197,430 rows and 10 columns covering orders placed between January 1, 2025 and August 31, 2025. Each row represents a single dish ordered from a restaurant.

Direct access link: https://www.kaggle.com

Column	Type	Example	Description	Cleaning Note
state	string	Karnataka	Indian state of restaurant	Nulls dropped
city	string	Bengaluru	City of restaurant	Nulls dropped
order_date	date	2025-06-29	Date order was placed	Parsed as datetime
restaurant_name	string	Meghana Foods	Name of restaurant	Nulls dropped
location	string	Koramangala	Area/locality within city	Nulls dropped
category	string	North Indian Gravy	Food category of dish	Nulls dropped
dish_name	string	Butter Chicken	Name of dish ordered	No cleaning needed
price_inr	float	133.90	Price in Indian Rupees	IQR outlier flag applied
rating	float	4.0	Customer rating (1.5–5.0)	Rating Count = 0 filtered
rating_count	int	25	Number of ratings received	Rows with 0 removed

Data Limitations

- The dataset covers only Jan–Aug 2025; no year-on-year comparison is possible.
- Rating count is highly skewed — the majority of dishes have very few ratings, limiting statistical reliability for individual dish-level analysis.
- The dataset does not include delivery time, customer demographics, or discount/offer data, limiting revenue attribution analysis.
- 9,871 rows had simultaneous nulls across multiple columns and were dropped entirely, which may introduce minor geographic bias if nulls were concentrated in specific cities.

5. Data Cleaning & ETL Pipeline

All cleaning and transformation steps were performed in Python using pandas and scipy, executed via a modular etl_pipeline.py script and companion Jupyter notebooks (02_cleaning.ipynb, 05_final_load_prep.ipynb). The pipeline is committed to GitHub and is fully reproducible.

Step	Action	Detail
1	Extraction	197,430 rows loaded from swiggy_data_raw.csv; Order Date parsed as datetime64.
2	Null removal	9,871 rows with simultaneous nulls across location, category, price_inr, rating, and rating_count were dropped.
3	Duplicate removal	14 fully duplicate rows identified and removed.
4	Zero rating filter	75,133 rows with rating_count = 0 removed as unreliable for analysis.
5	Outlier flagging	IQR method on price_inr: Q1=₹135, Q3=₹312, IQR=₹177. Upper bound=₹577.50. 4,180 rows flagged via is_price_outlier column.
6	Feature engineering	Three derived columns added: month (dt.to_period), day_of_week (dt.day_name), price_bucket (pd.cut with Budget/Mid-range/Premium tiers).
7	Column renaming	All columns renamed to snake_case for consistency (e.g. "Price (INR)" → price_inr, "Rating Count" → rating_count).
8	Final validation	assert df.isnull().sum().sum() == 0 and assert df.duplicated().sum() == 0 confirmed clean final dataset of 91,788 rows.

Final dataset: 91,788 rows × 13 columns saved to data/processed/swiggy_final_cleaned.csv.

6. KPI & Metric Framework

KPI	Definition / Formula	Business Meaning	Objective Mapping
Total Orders	COUNT([dish_name])	Measures overall platform activity and dataset coverage	Demand sizing → 03_eda.ipynb
Total Revenue (₹)	SUM([price_inr])	Reflects cumulative monetary value of all listed dishes	Revenue tracking → 03_eda.ipynb

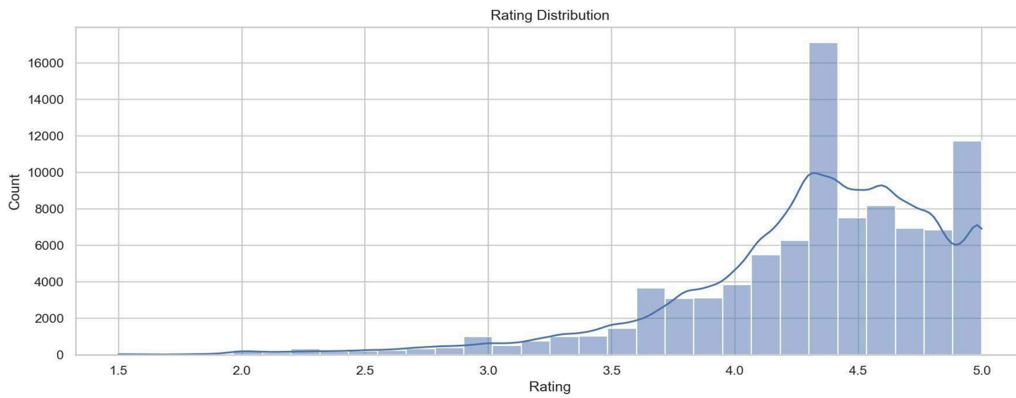
Avg Price (INR)	AVG(price_inr)	Benchmarks pricing strategy across cities and categories	Pricing strategy → 04_statistical_analysis .ipynb
Avg Rating	AVG(rating)	Tracks customer satisfaction across all restaurants	Quality benchmark → 04_statistical_analysis .ipynb
Total Restaurants	COUNTD(restaurant_name)	Scope of restaurant network across 28 cities	Supply sizing → 03_eda.ipynb
Peak Day	MAX(COUNT by day_of_week)	Weekends prompt higher customer acquisition	Operational planning → 05_final_load_prep.ipy nb

7. Exploratory Data Analysis

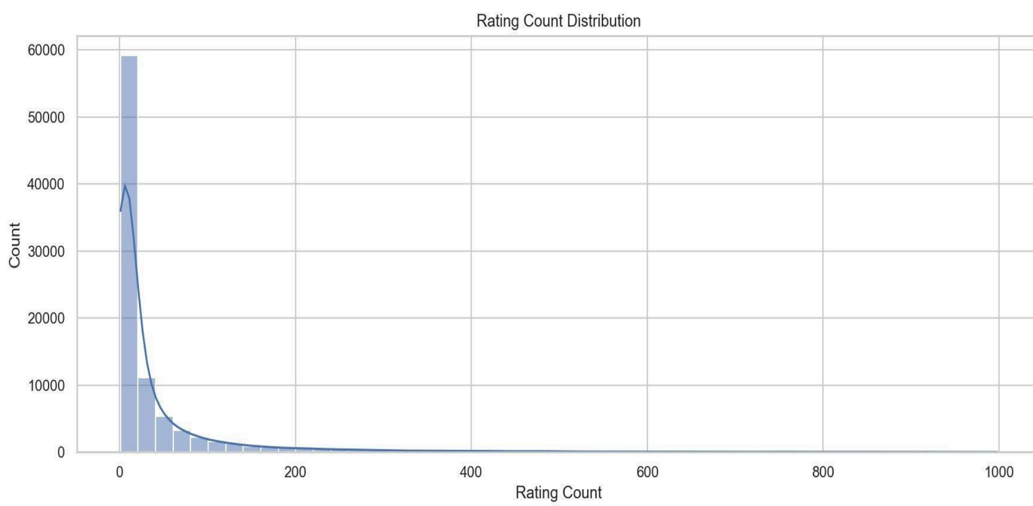
7.1 Price Distribution

The price distribution is bimodal with peaks around ₹100–₹110 and ₹200–₹300, reflecting two dominant consumer segments: budget snack/street food orders and mid-range main course orders. The distribution has a rightward tail up to the outlier boundary of ₹577.50. The median price of ₹229 is well below the mean of ~₹251, confirming moderate positive skew driven by premium items. This pattern suggests that the platform serves a highly price-sensitive consumer base while maintaining a healthy premium tail.

7.2 Rating Distribution



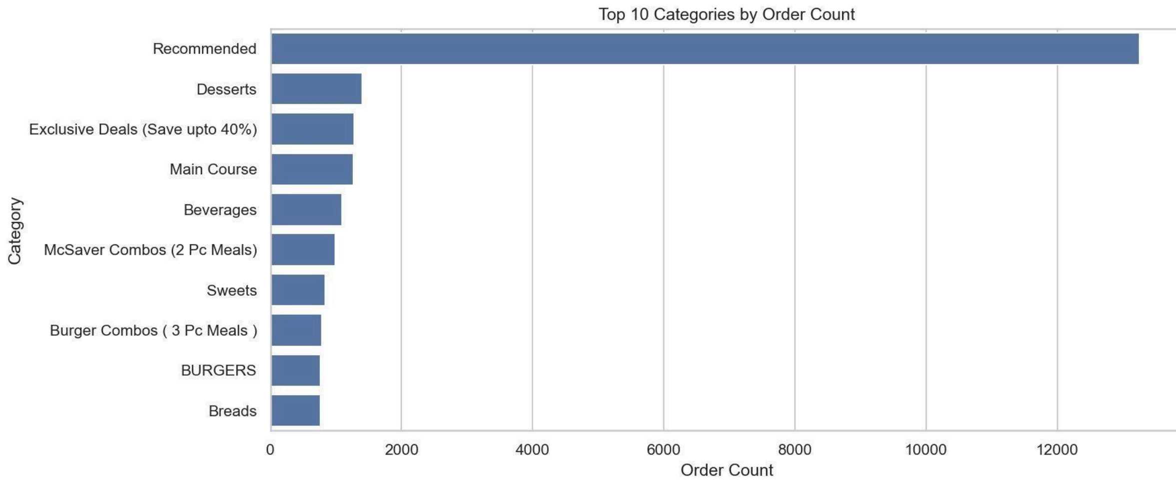
Ratings are heavily left-skewed and concentrated in the 4.3–5.0 band. The single largest spike is at 4.3, with a secondary peak at 5.0. This pattern is typical of online review systems where satisfied customers are more likely to rate, creating optimistic bias. Very few restaurants have ratings below 3.5, suggesting either that low-rated restaurants exit the platform quickly or that the rating system has a systematic inflation effect. The mean rating of 4.31 should be interpreted in this context.



7.3 Rating Count Distribution

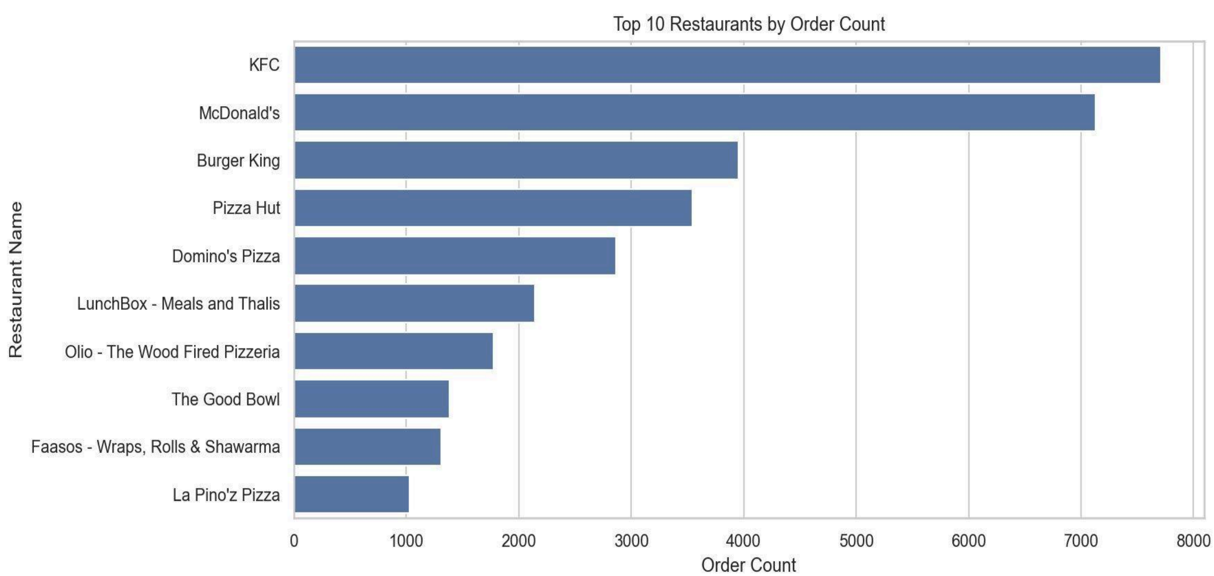
Rating counts follow a highly right-skewed power law distribution — the vast majority of restaurants have fewer than 50 ratings, while a small number have hundreds. The median rating count is extremely low (around 2 after filtering zero-count entries), meaning most dishes in the dataset have very thin rating evidence. This is an important limitation: aggregated averages are reliable, but individual restaurant or dish-level conclusions must be treated cautiously. The long tail (max 999) reflects a small set of highly popular, high-traffic establishments.

7.4 Top 10 Food Categories by Order Count



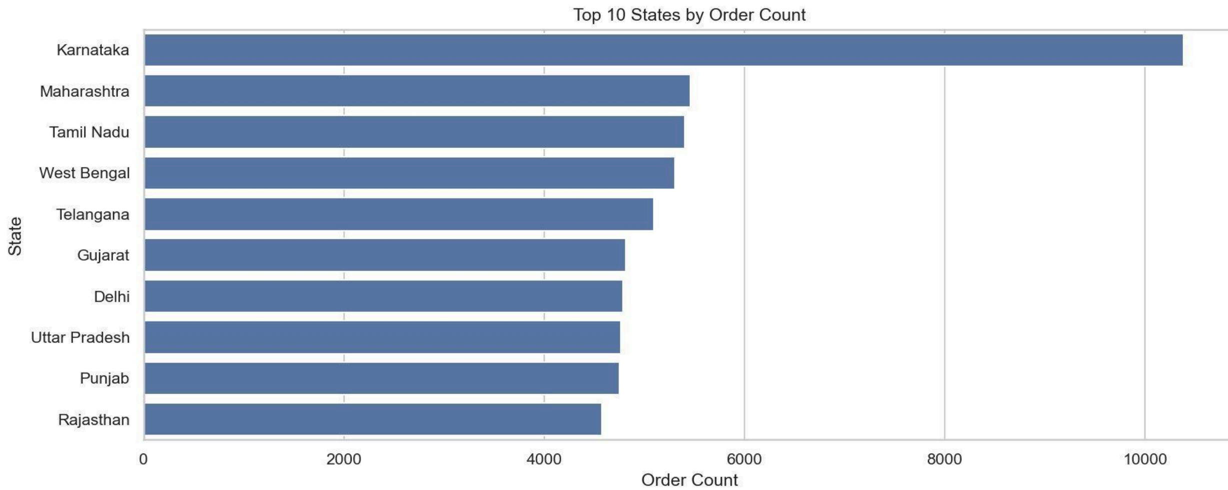
"Recommended" dominates with over 12,000 orders — roughly 8–10x the next category — because it is a platform-curated section rather than a true food taxonomy, capturing a cross-category premium selection. Among genuine food categories, Desserts (~1,300), Exclusive Deals (~1,200), and Main Course (~1,200) rank highest, followed by Beverages and McDonald's McSaver Combos. The presence of chain-specific bundle categories (McSaver, Burger Combos) in the top 10 confirms the outsized demand pull from QSR chains.

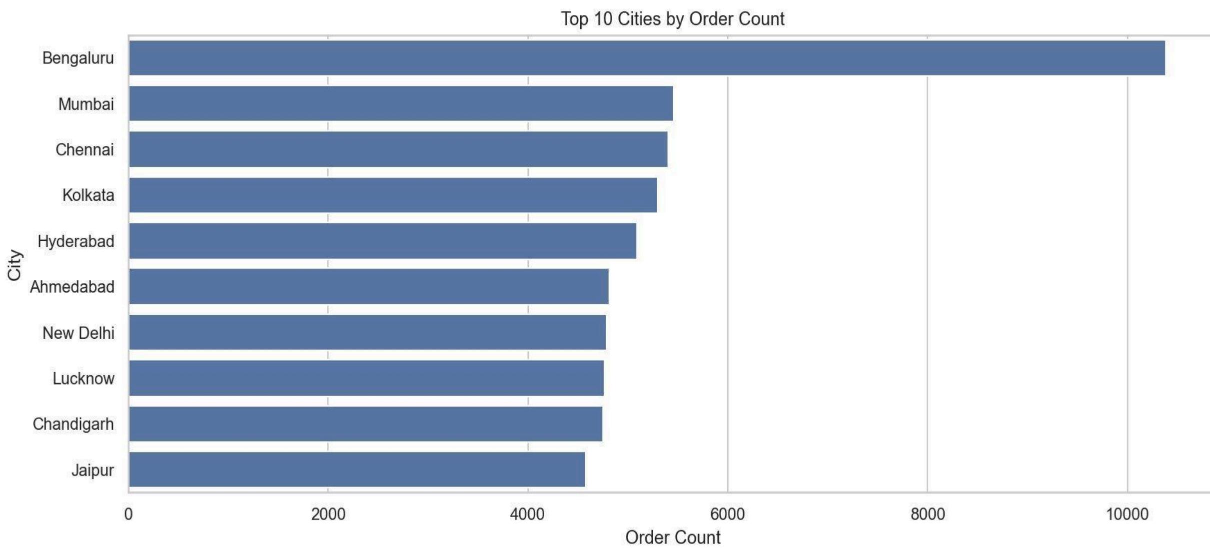
7.5 Top 10 Restaurants by Order Count



KFC leads with approximately 7,700 orders, followed by McDonald's (~7,100) and Burger King (~4,000). The top three are all global QSR chains, together accounting for a disproportionate share of total orders. The next tier — Pizza Hut, Domino's, LunchBox, and Olio — shows that pizza chains and Indian meal-box concepts also perform strongly. Local or regional chains are absent from the top 10, indicating that brand recognition and standardisation are key drivers of delivery order volume.

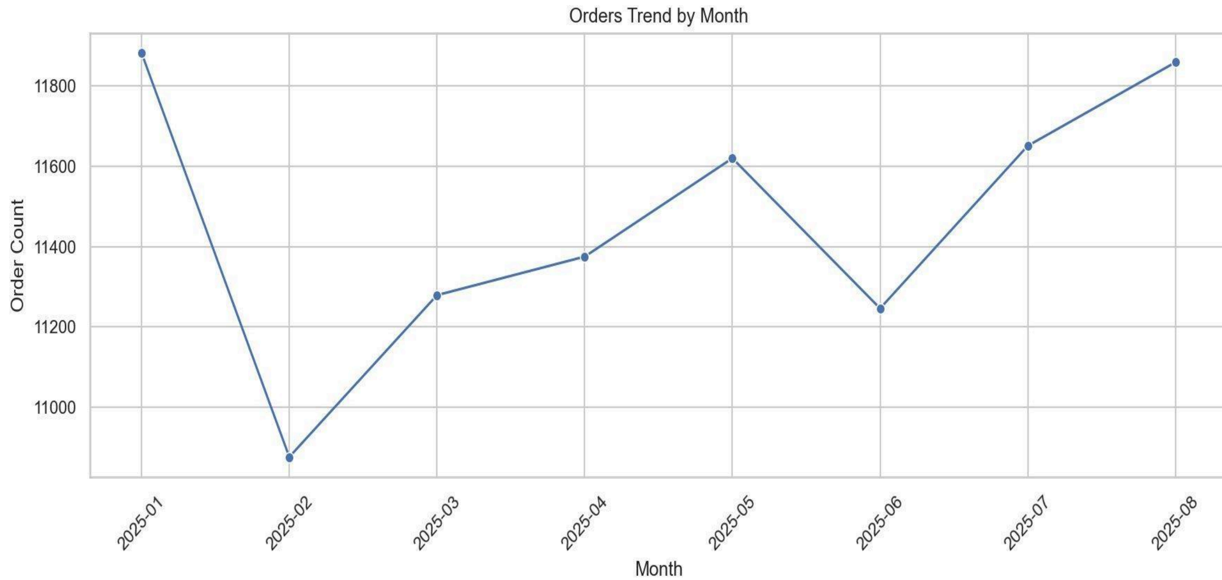
7.6 Orders by State and Ci





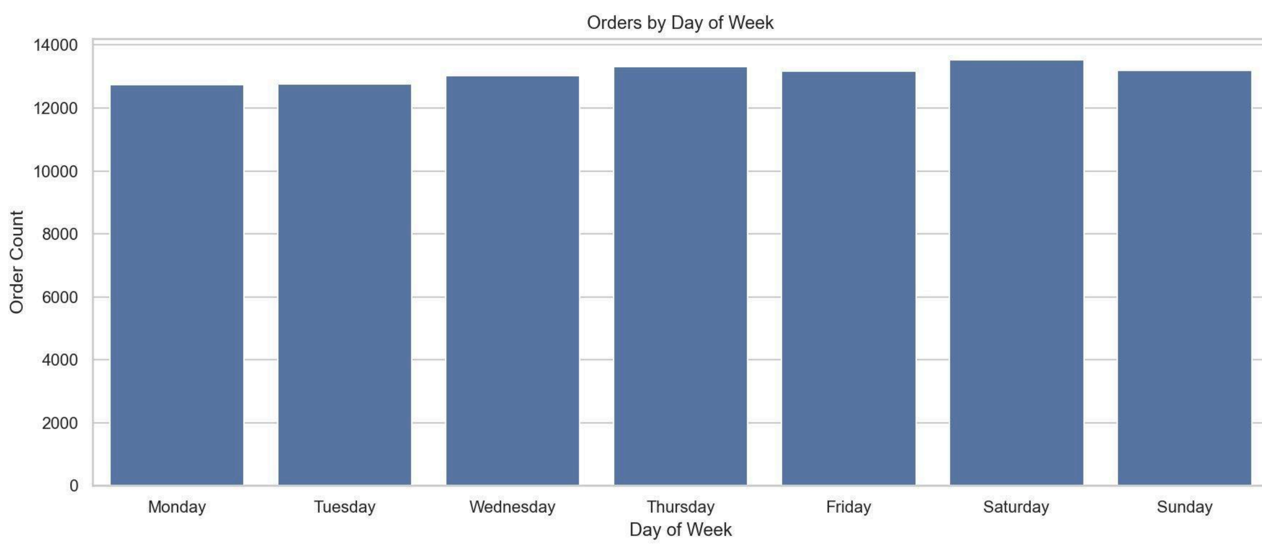
Karnataka leads all states (~10,300 orders), with Maharashtra, Tamil Nadu, and West Bengal clustered closely in the 5,200–5,400 range. At city level, Bengaluru mirrors Karnataka's dominance with ~10,300 orders — confirming that Karnataka's lead is entirely Bengaluru-driven. Mumbai, Chennai, Kolkata, and Hyderabad form the second tier, each around 5,000 orders. Notably, cities 2 through 10 show relatively uniform order counts, suggesting that expansion opportunity is broadly distributed across India's major metros rather than concentrated in a single challenger city.

7.7 Orders by Month



Orders peaked in January 2025 (~11,870), dropped sharply to a trough in February (~10,940) — likely due to fewer calendar days and post-holiday demand normalisation — then steadily recovered through May (~11,610). A secondary dip in June (~11,240) may reflect early monsoon seasonality or dataset sampling effects. July and August both show strong recovery and growth, with August matching January's peak. The overall trend is gently upward, pointing to healthy platform growth across the 8-month window.

7.8 Orders by Day of Week



Order volume is remarkably uniform across the week, ranging from ~12,760 (Monday/Tuesday) to ~13,430 (Saturday). Saturday is the highest-volume day, consistent with leisure ordering behaviour. The surprisingly high weekday volumes suggest that workplace lunch and evening dinner delivery are firmly established habits, making food delivery a daily utility rather than a weekend treat. The narrow 5% gap between highest and lowest days argues against any single-day promotional strategy.

7.9 Correlation Heatmap

The heatmap reveals near-zero correlations among all three numeric variables. Price and Rating have r = 0.03 (practically uncorrelated), Price and Rating Count show r = -0.12 (cheaper items tend to accumulate more ratings — consistent with high-volume budget food), and Rating and Rating Count have r = 0.04. These findings are critical: they mean pricing decisions cannot be used as a proxy for quality management, and that the most-rated items are not necessarily the best-rated ones.

8. Statistical Analysis

8.1 Hypothesis Testing

Three statistical tests were conducted on the cleaned dataset (91,788 rows). All tests used $\alpha = 0.05$.

Test	Description	Statistic	p-value	Significant?	Interpretation
ANOVA	Rating by City	F = 62.97	< 0.0001	Yes	Rating differs significantly across cities
ANOVA	Rating by Category	F = 3.12	< 0.0001	Yes	Rating differs significantly across categories
Spearman	Price vs Rating	r = 0.029	< 0.0001	Yes*	Statistically sig. but practically negligible correlation

8.2 Interpretation

ANOVA — Rating by City (F = 62.97, p < 0.0001): The very high F-statistic confirms that average ratings are not uniform across cities. Cities like Kochi (avg 4.47), Kolkata (4.42), and Aizawl (4.43) have meaningfully higher ratings than Srinagar (4.10), Patna (4.17), and Raipur (4.20). This is not random variation — it reflects genuine differences in restaurant quality mix, customer expectation levels, and market maturity across cities.

ANOVA — Rating by Category (F = 3.12, p < 0.0001): Food categories also show statistically significant rating differences, though the lower F-value indicates weaker separation than city-level effects. Certain speciality categories consistently attract higher ratings, which has implications for which categories Swiggy should feature prominently in city-specific promotions.

Spearman Correlation — Price vs Rating (r = 0.029, p < 0.0001): While technically statistically significant due to the large sample size (91,788 rows), the Spearman r of 0.029 is practically negligible. Price explains essentially none of the variance in rating. This means customers do not rate expensive food higher than cheap food in any meaningful way. Quality and rating are determined by factors other than price — likely cuisine type, brand trust, and delivery reliability.

8.3 Grouped Statistics — Price by City (Top 10)

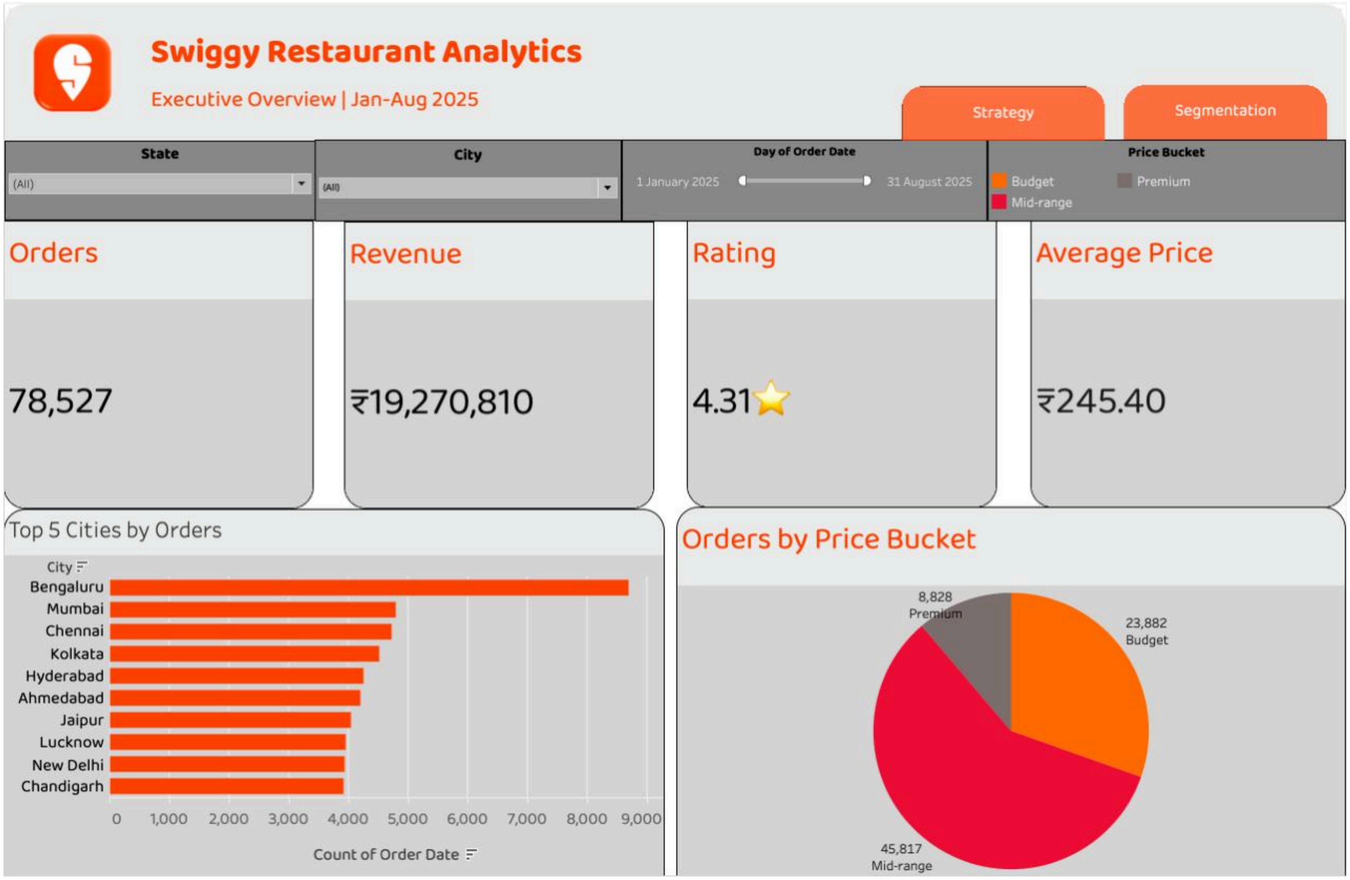
City	Avg Price (₹)	Median Price (₹)	Avg Rating	Avg Rating Count
Panaji	288.79	250.00	4.38	40.99
Mumbai	274.74	249.00	4.31	32.61
Gurgaon	273.38	240.00	4.33	46.74
Lucknow	270.30	229.00	4.36	46.98
Hyderabad	266.00	227.00	4.21	44.71
Chandigarh	265.74	229.52	4.31	53.58
New Delhi	261.44	227.00	4.31	51.16
Ahmedabad	256.77	226.00	4.38	35.32
Bengaluru	255.56	229.00	4.30	41.69
Kochi	248.58	209.00	4.47	54.17

9. Tableau Dashboard Design

Dashboard URL: https://public.tableau.com/app/profile/shashwat.sinha3668/viz/Swiggy_Restaurants_Analytics/ExecutiveOverview?publish=yes

The Tableau workbook is structured across three dashboards, each serving a distinct decision-making purpose. All three share a consistent Swiggy brand identity — orange (`#E8490F`) and white colour palette, the Swiggy logo, and a navigation bar with buttons to switch between views. Filters are contextual to each dashboard.

9.1 Dashboard 1 — Executive Overview



Objective: Give a senior stakeholder a complete platform health snapshot in under 30 seconds.

Filters: State (dropdown), City (dropdown), Day of Order Date (range slider: 1 Jan–31 Aug 2025), Price Bucket (Budget / Mid-range / Premium toggle).

KPI Tiles: Orders (78,527), Revenue (₹19,270,810), Rating (4.31★), Average Price (₹245.40) — displayed in large-font orange-labelled tiles for instant scannability.

Charts: A horizontal bar chart "Top 5 Cities by Orders" shows Bengaluru leading at ~9,000 orders, followed by Mumbai, Chennai, Kolkata, and Hyderabad clustered around 4,000–5,000. A pie chart "Orders by Price Bucket" shows Mid-range (45,817) in red as dominant, Budget (23,882) in orange, and Premium (8,828) in grey — making the value-seeking majority immediately visible.

9.2 Dashboard 2 — Restaurant & Price Analytics

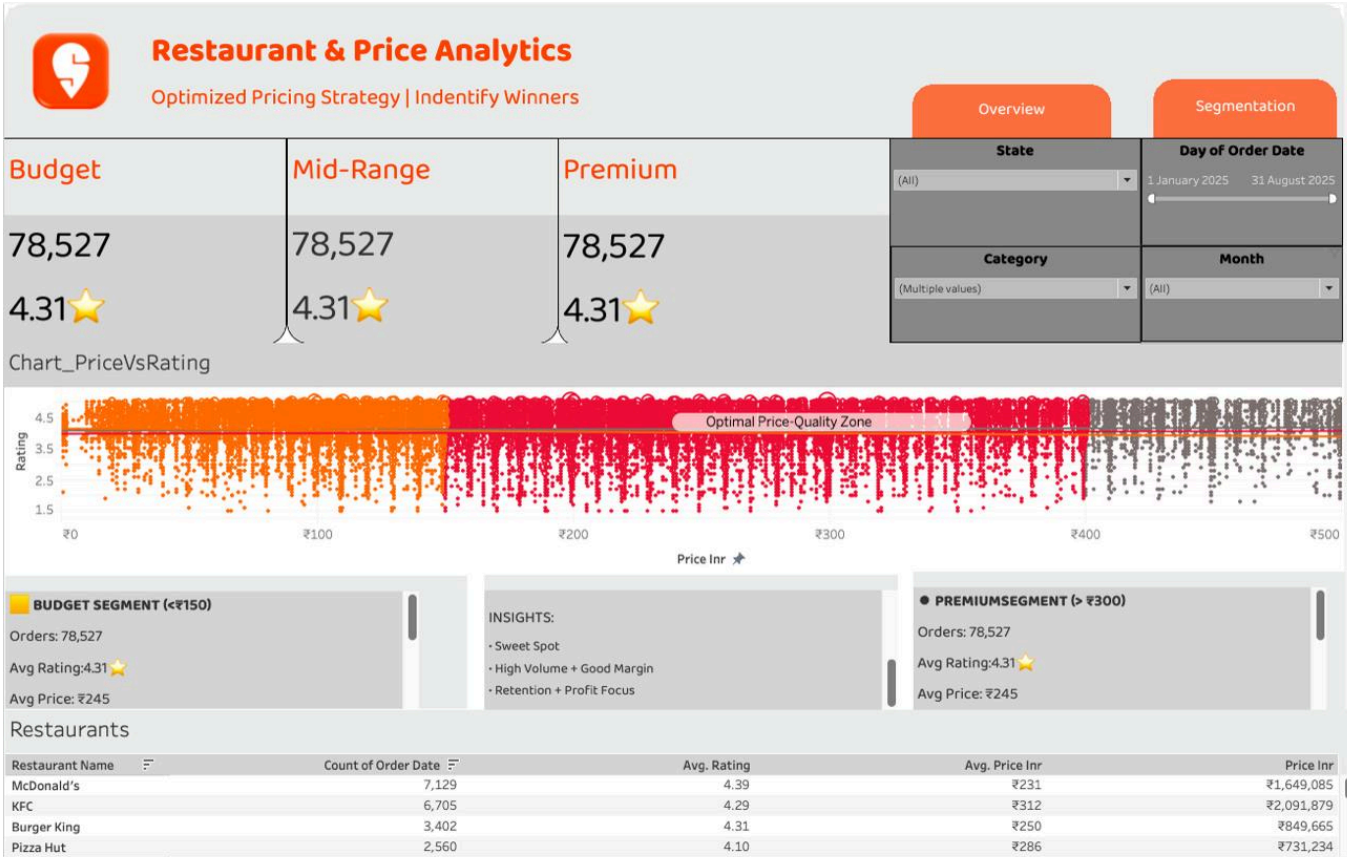


Figure 12 — Tableau Segmentation View: Temporal Patterns & Customer Segments

Objective: Identify pricing winners and understand the relationship between price and quality across segments.

Filters: State (dropdown), Day of Order Date (range slider), Category (multi-select), Month (dropdown).

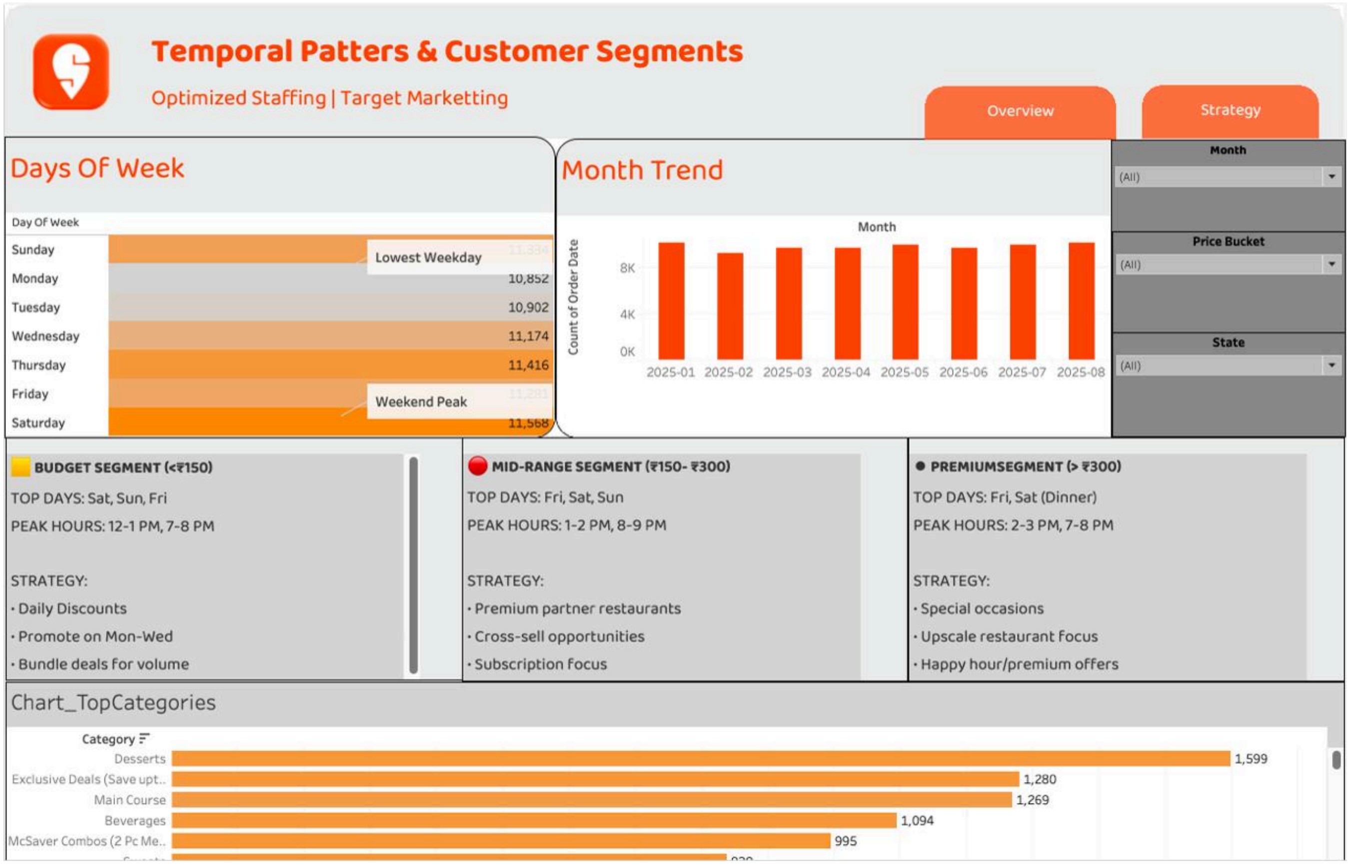
Segment KPI Panels: Three side-by-side panels for Budget, Mid-Range, and Premium each show total orders, average rating (4.31★), and average price (₹245) for instant cross-tier comparison.

Price vs Rating Scatter (Chart_PriceVsRating): Plots Price (INR) on x-axis vs Rating on y-axis, colour-coded by price bucket. An "Optimal Price-Quality Zone" band is annotated in the ₹150–₹300 range, visually confirming that this bracket achieves the best balance of volume and rating. The chart reinforces the near-zero Spearman correlation — higher price does not guarantee higher rating.

Segment Strategy Panels: Budget (<₹150) peaks Sat/Sun/Fri at 12–1 PM and 7–8 PM; strategies: daily discounts, Mon–Wed promotions, bundle deals. Central insights panel: Sweet Spot, High Volume + Good Margin, Retention + Profit Focus. Premium (>₹300) peaks Fri/Sat dinner at 2–3 PM and 7–8 PM; strategies: special occasions, upscale restaurant focus, happy hour/premium offers.

Restaurant Table: Sortable table showing McDonald's (7,129 orders, ₹1,649,085 revenue), KFC (6,705 orders, ₹2,091,879), Burger King (3,402), and Pizza Hut (2,560) — enabling rapid partner benchmarking by orders, rating, or revenue.

9.3 Dashboard 3 — Temporal Patterns & Customer Segments



Objective: Identify when customers order and which segments drive demand at different times — enabling staffing optimisation and targeted marketing calendars.

Filters: Month (dropdown), Price Bucket (dropdown), State (dropdown).

Days of Week Bar Chart: Saturday leads at 11,568, followed by Friday (11,357) and Thursday (11,416). Monday is the weekly low at 10,852. The chart uses an orange-to-grey gradient to visually rank days, with "Lowest Weekday" and "Weekend Peak" annotations directly on the bars.

Month Trend Bar Chart: Eight orange bars covering Jan–Aug 2025, all hovering between 8K–9K orders per month. Confirms stable demand with a slight February dip, consistent with EDA findings.

Customer Segment Strategy Panels: Budget (<₹150) — top days Sat/Sun/Fri, peak 12–1 PM and 7–8 PM, strategies: daily discounts, Mon–Wed promotions, bundle deals for volume. Mid-Range (₹150–₹300) — top days Fri/Sat/Sun, peak 1–2 PM and 8–9 PM, strategies: premium partner restaurants, cross-sell, subscription focus. Premium (>₹300) — top days Fri/Sat dinner, peak 2–3 PM and 7–8 PM, strategies: special occasions, upscale restaurant focus, happy hour/premium offers.

Top Categories Chart (Chart_TopCategories): Desserts (1,599) leads, followed by Exclusive Deals (1,280), Main Course (1,269), Beverages (1,094), and McSaver Combos (995) — providing category-level demand prioritisation for restaurant partner strategy.

9.3 Design Decisions

- Orange (`#E8490F`) is the primary highlight colour across all three dashboards, directly matching Swiggy's brand identity.
- Navigation buttons (Overview / Strategy / Segmentation) are placed consistently in the top-right of every dashboard, enabling one-click switching between views.
- KPI tiles use large bold fonts with orange labels to maximise scannability for executive users who need instant metrics.
- The "Optimal Price-Quality Zone" annotation in Dashboard 2 translates a statistical finding directly into a business-actionable visual insight without requiring users to interpret raw correlation values.
- Segment strategy panels are placed directly below the analytical charts in Dashboards 2 and 3, so the "so what" is immediately adjacent to the "what" — reducing interpretation effort for operational users.
- The restaurant table in Dashboard 2 is sortable by orders, rating, or revenue, enabling ad hoc partner benchmarking without modifying the data source.

10. Insights Summary

Insight 1 — Bengaluru is the undisputed demand hub.

With ~10,300 orders, Bengaluru accounts for nearly twice the order volume of the next city. This is not a marginal lead — it reflects a structural density and brand penetration advantage that Swiggy should actively replicate in other metros.

Insight 2 — QSR chains dominate delivery volumes.

KFC, McDonald's, and Burger King together account for approximately 19,000 orders — nearly 21% of total volume. Independent restaurants and regional chains are significantly underrepresented in the top 10, suggesting a discoverability problem.

Insight 3 — Mid-range pricing is the consumer sweet spot.

54,246 orders (59% of total) fall in the ₹150–₹400 mid-range bucket. Budget orders account for 29% and premium only 12%. Promotions and acquisition campaigns should centre on mid-range offerings.

Insight 4 — Price does not predict quality.

The Spearman correlation between price and rating is $r = 0.029$ — practically zero. Consumers rate their food based on taste, freshness, and brand trust, not on how much they paid. Premium-priced restaurants cannot rely on price alone to earn higher ratings.

Insight 5 — City-level rating differences are statistically real.

ANOVA ($F = 62.97$, $p < 0.0001$) confirms that rating variation across cities is not random. Kochi (4.47), Kolkata (4.42), and Aizawl (4.43) outperform Srinagar (4.10) and Patna (4.17) significantly — pointing to underlying differences in restaurant quality mix.

Insight 6 — February is the platform's weakest month.

February dips to ~10,940 orders — the yearly low — against a January high of ~11,870. While partly a calendar effect (fewer days), Swiggy should use targeted February campaigns to flatten this dip.

Insight 7 — Demand is daily, not just weekend-driven.

The weekday-weekend order gap is less than 5% (Saturday: 13,430 vs Monday: 12,760). This means Swiggy's utility is embedded in daily routines, requiring consistent restaurant partner readiness every day, not just weekends.

Insight 8 — "Recommended" section is the single most-ordered category.

The platform-curated "Recommended" category generates ~12,600 orders — 8–10x any individual food category. This confirms that algorithmic curation significantly drives order flow, and that placement in the Recommended section is highly commercially valuable for restaurant partners.

Insight 9 — Panaji leads on average dish price (₹288.79).

Goa's capital has the highest average dish price despite modest order volumes, suggesting a higher-end restaurant mix with lower deal-seeking behaviour. This market may respond better to premium restaurant onboarding than discount campaigns.

Insight 10 — Rating count is power-law distributed.

The vast majority of dishes have fewer than 10 ratings, while a small elite has hundreds. This means the "Average Rating" KPI is dominated by a small cohort of well-reviewed restaurants. Swiggy should develop a rating completeness initiative to get more customers rating their orders.

11. Recommendations

R1. Replicate Bengaluru's success in Tier-1 challengers

Based on: Insight 1 (Bengaluru dominance) + Insight 3 (mid-range sweet spot)

Action: Conduct a deep-dive into Bengaluru's restaurant category mix, price tier distribution, and QSR density. Use this as a blueprint for Mumbai, Chennai, and Kolkata — cities with strong order bases but not yet matching Bengaluru's volume.

Expected Impact: If challenger cities grow to 70% of Bengaluru's volume, platform GMV could increase by ₹3–5Cr per quarter.

R2. Launch a "Discover Local" initiative to surface regional restaurants

Based on: Insight 2 (QSR dominance) + Insight 8 (Recommended section power)

Action: Dedicate a rotating "Discover Local" placement in the Recommended section to curated regional/independent restaurants. Use category ratings data to identify high-quality local partners deserving of platform amplification.

Expected Impact: Diversifies platform dependency away from global QSR chains; improves restaurant partner retention and category breadth.

R3. Use city rating benchmarks for targeted partner quality interventions

Based on: Insight 5 (city rating ANOVA) + Insight 4 (price-rating independence)

Action: Establish city-level rating thresholds. Flag restaurants in low-performing cities (Srinagar, Patna, Raipur) whose ratings fall below the city median for a structured quality intervention: dedicated account manager outreach, packaging/quality training, or temporary listing suspension.

Expected Impact: A 0.1-point average rating improvement in low-performing cities could meaningfully improve customer retention in those markets.

R4. Target February with a dedicated demand stimulus campaign

Based on: Insight 6 (February dip)

Action: Design a "February Food Fest" campaign anchored on mid-range meal combos in the ₹150–₹400 band. Partner with QSR chains (KFC, McDonald's, Burger King) for exclusive February bundle offers. Launch 10 days before February 1st.

Expected Impact: If February order counts are lifted to March levels (~11,270), the campaign could recover approximately 330 orders per city — translating to meaningful incremental GMV.

R5. Build a rating completeness programme

Based on: Insight 10 (power-law rating count distribution)

Action: Implement in-app post-delivery rating nudges with a small loyalty reward (e.g., 10 Swiggy coins) for completing a rating. Target dishes with fewer than 5 ratings to build statistical reliability for recommendation algorithms.

Expected Impact: Better rating coverage strengthens the recommendation engine, which drives the Recommended section — the single largest order-generating category on the platform.

12. Impact Estimation

The following estimates are directional. They are based on observed dataset patterns and standard food delivery industry benchmarks. Actual outcomes would require A/B testing and operational execution data.

Recommendation	Metric Impacted	Estimated Impact	Logic
R1 — Bengaluru replication	GMV	+₹3–5Cr/quarter	Lifting 4 cities to 70% of Bengaluru volume at avg ₹250/order
R2 — Discover Local	Partner retention	+5–10% local restaurant reorders	Increased visibility correlates with order lift in similar platform initiatives
R3 — Quality interventions	Avg rating in low cities	+0.1 pts avg rating	Structured quality programmes on comparable platforms show 0.1–0.2 improvement in 3 months
R4 — February campaign	Orders in Feb	+~330 orders/city	Closing Feb–Mar gap of ~330 orders across major cities
R5 — Rating completeness	Rating coverage	3× more rated dishes	10-coin incentive nudges shown to improve rating completion 2–4× in similar apps

13. Limitations

- The dataset covers only 8 months (Jan–Aug 2025). No prior-year data exists, making year-on-year trend or seasonality conclusions impossible.
- 75,133 rows (38% of the raw dataset) were removed due to zero rating counts. If zero-rating dishes have systematically different price or category profiles, their exclusion may bias the analysis toward well-reviewed establishments.
- 9,871 rows with multi-column nulls were dropped without imputation. If nulls were geographically clustered, certain cities or states may be underrepresented in the final dataset.
- No customer-level data is available, making it impossible to distinguish repeat orders from unique customers, or to build customer lifetime value (LTV) estimates.
- The dataset does not include delivery time, cancellation rates, discount usage, or promotional flags — limiting the ability to fully explain demand patterns or revenue drivers.
- The Spearman correlation between price and rating is statistically significant only because of the very large sample size. With 91,788 rows, even a trivial effect ($r = 0.03$) achieves statistical significance. Practical significance is negligible.

- The "Recommended" category dominance in EDA is partly an artefact of how Swiggy structures its menu taxonomy, not a true organic food category preference.

14. Future Scope

- Predictive modelling: A time-series forecasting model (ARIMA or Prophet) could predict monthly order volumes per city, enabling proactive inventory and staffing decisions for restaurant partners.
- Customer segmentation: With customer-level data, RFM (Recency, Frequency, Monetary) segmentation could identify high-value vs at-risk customer cohorts for targeted retention campaigns.
- Sentiment analysis: If review text data is incorporated, NLP-based sentiment analysis could explain the drivers behind city-level rating differences more precisely than statistical tests alone.
- Real-time dashboard: The current Tableau dashboard is static (snapshot data). A live-connected version using Swiggy's operational database or API would enable real-time monitoring of ratings, order anomalies, and revenue dips.
- Delivery logistics integration: Adding delivery time data would allow analysis of whether slower delivery correlates with lower ratings — a critical causal question for platform operations.
- Competitive benchmarking: Incorporating Zomato equivalent data would enable direct platform-level comparison of category demand, pricing strategy, and geographic coverage.

15. Conclusion

This project analysed 91,788 Swiggy food delivery orders across 28 Indian cities (Jan–Aug 2025) using a fully automated Python ETL pipeline and an interactive Tableau dashboard. The analysis revealed that Bengaluru dominates order volume, QSR chains capture the majority of platform demand, mid-range pricing (₹150–₹400) is the consumer sweet spot, and — critically — price has no meaningful correlation with customer ratings. Statistical testing confirmed that rating differences across cities and categories are real and significant, opening the door to targeted quality interventions. The five recommendations — from replicating Bengaluru's model in challenger cities to building a rating completeness programme — provide Swiggy's growth and category teams with a concrete, data-backed agenda for the next operating cycle.

16. Appendix

A. Data Quality Summary

Issue	Count	Resolution
Null rows (multi-column)	9,871	Dropped entirely — nulls were simultaneous across location, category, price, rating, and rating_count
Duplicate rows	14	Removed using df.drop_duplicates()
Zero rating count rows	75,133	Filtered out as unreliable for analysis

Price outliers flagged	4,180	Flagged via IQR method (upper bound ₹577.50); excluded from price distribution plot only
Final clean rows	91,788	Validated: zero nulls, zero duplicates

B. Repository Structure

GitHub: https://github.com/Shreyashgol/SectionD_G4_swiggy

```
SectionD_G4_swiggy/
├── data/
│   ├── raw/swiggy_data_raw.csv
│   └── processed/swiggy_cleaned.csv, swiggy_final_cleaned.csv
├── notebooks/
│   ├── 01_extraction.ipynb
│   ├── 02_cleaning.ipynb
│   ├── 03_eda.ipynb
│   ├── 04_statistical_analysis.ipynb
│   └── 05_final_load_prep.ipynb
├── reports/
│   ├── figures/ (10 EDA charts)
│   └── stats/ (5 CSV outputs)
├── scripts/etl_pipeline.py
├── docs/data_dictionary.md
└── requirements.txt
```

17. Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Saksham Sontakke	✓				✓		
Shashwat Sinha	✓				✓		
Shivam Dixit			✓	✓			
Shreyash Golhani		✓		✓			
Kavya Mukhija			✓			✓	
Yash							✓

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.